

## Problem Set 5

1. Compute  $\beta$ -function for the gauge coupling of Yang-Mills coupled to a complex scalar field in represent.  $R$  at 1-loop.

Compute it again for Yang-Mills coupled to  $N_f$  Dirac Fermions in rep.  $R_i$  &  $N_s$  complex scalars in  $R'_j$ .

2. Write down an action  
Noncommutative Yang-Mills theory  
on a lattice as a sum over  
plaquettes.

What's the path-integral measure?

Compute the Haar integrals

$$\int dU U_{ij}$$

$$\int dU U_{ij} U_{kl}$$

$$\int dU U_{ij} U_{kl}^*$$

and use them to argue that the vacuum expectation of Wilson loop operators  $W_C$  at infinite coupling is

$$\langle 0 | W_C | 0 \rangle \sim (1/g^2)^{A/a^2}$$

where  $A$  is the area of the Wilson loop &  $a$  is the lattice spacing.